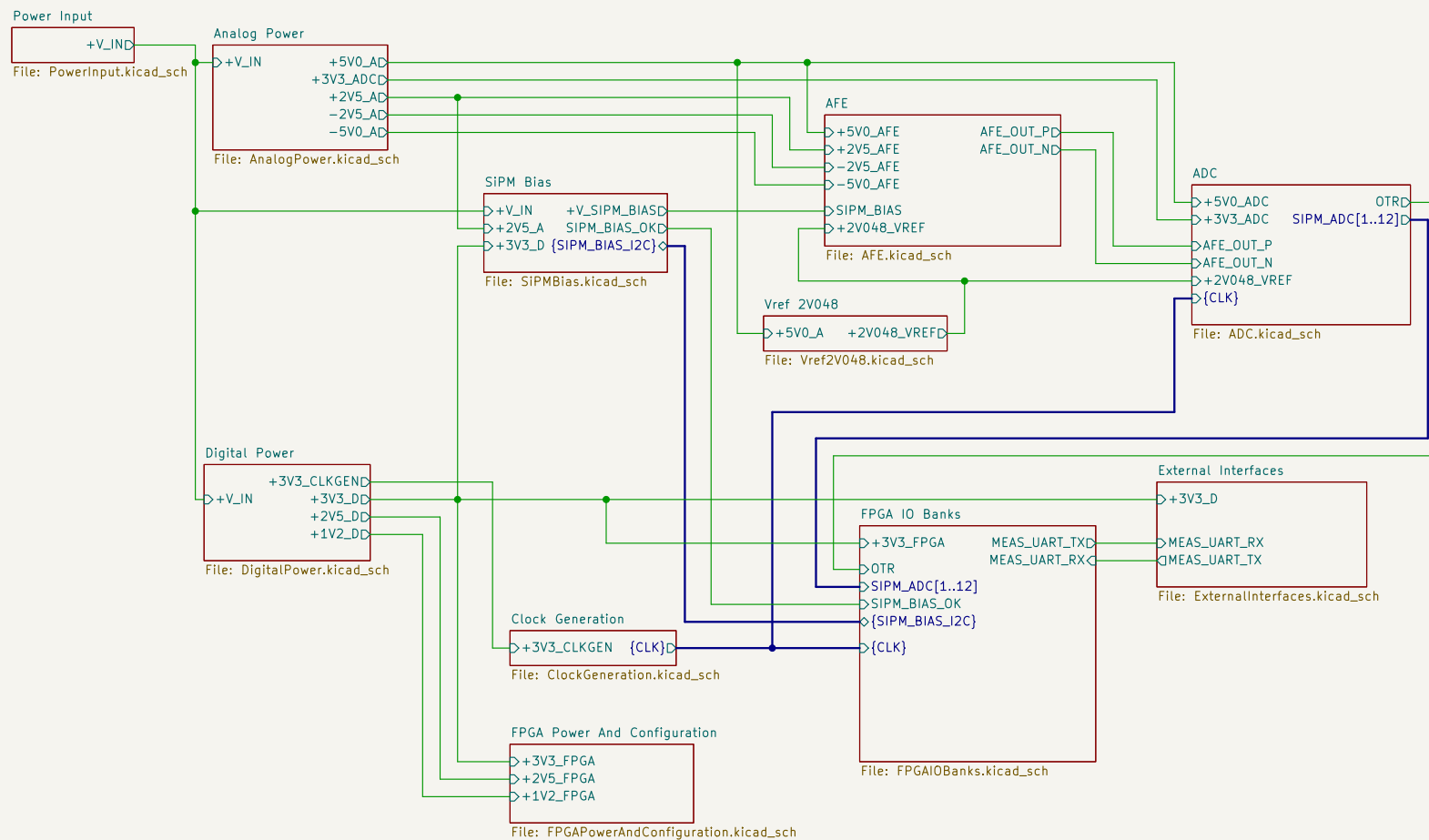
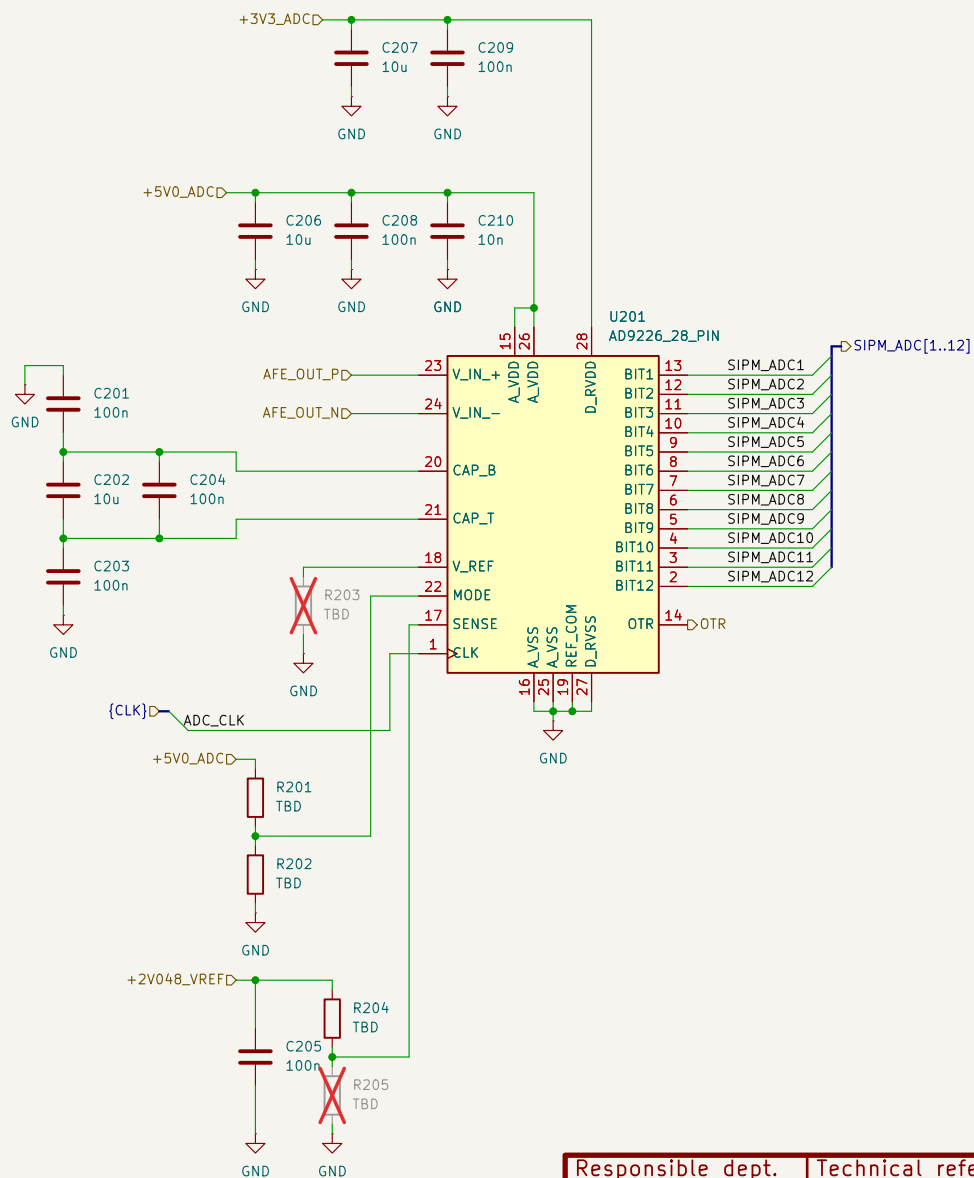
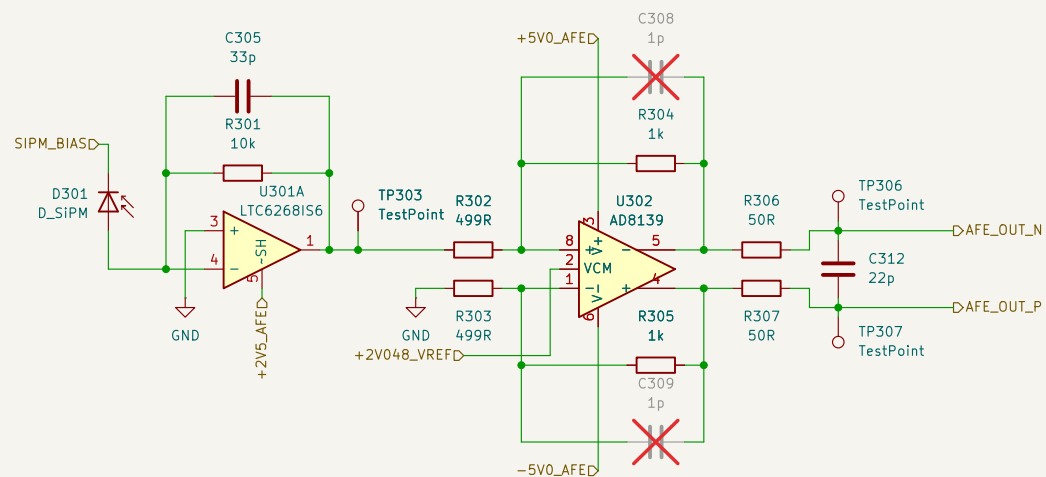




Responsible dept.		Technical reference		Document type		Document status	
				Schematic Diagram		DRAFT	
RG		Created by		Title, Supplementary title			
		Approved by		Top-Level System Architecture		Rev. 0.1	Date of issue 2026-04-11
						Lang.	Sheet 1/19

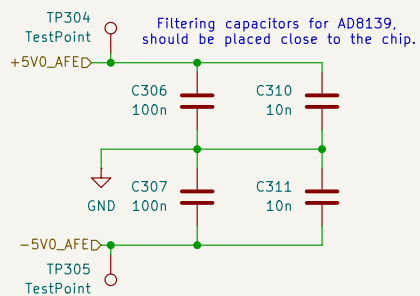
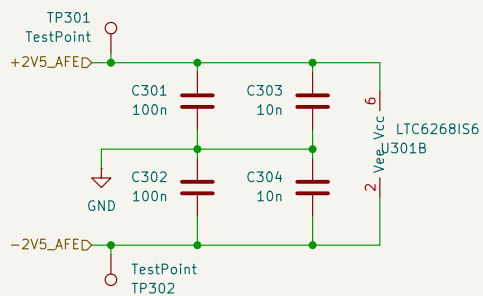


Responsible dept.	Technical reference	Document type	Document status
		Schematic Diagram	DRAFT
RG	Created by	Title, Supplementary title 12-bit 65MSPS Analog-to-Digital Converter	Rev. 0.1
	Approved by		
			Date of issue 2026-04-11
			Lang.
			Sheet 2/19

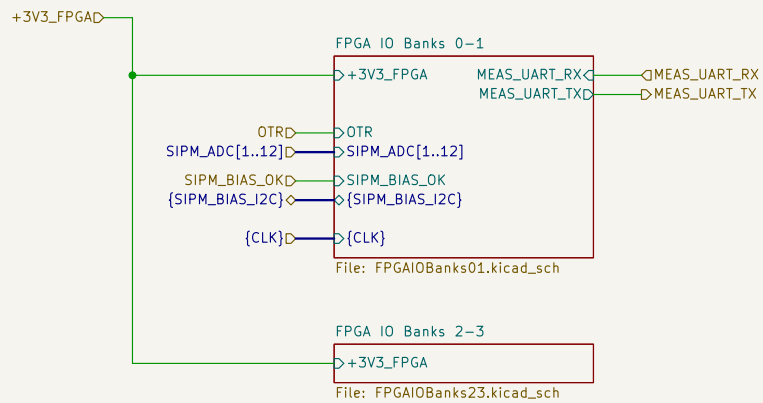


All capacitors (except for voltage filtering) are COG / NP0.
All resistors are 1% or better.

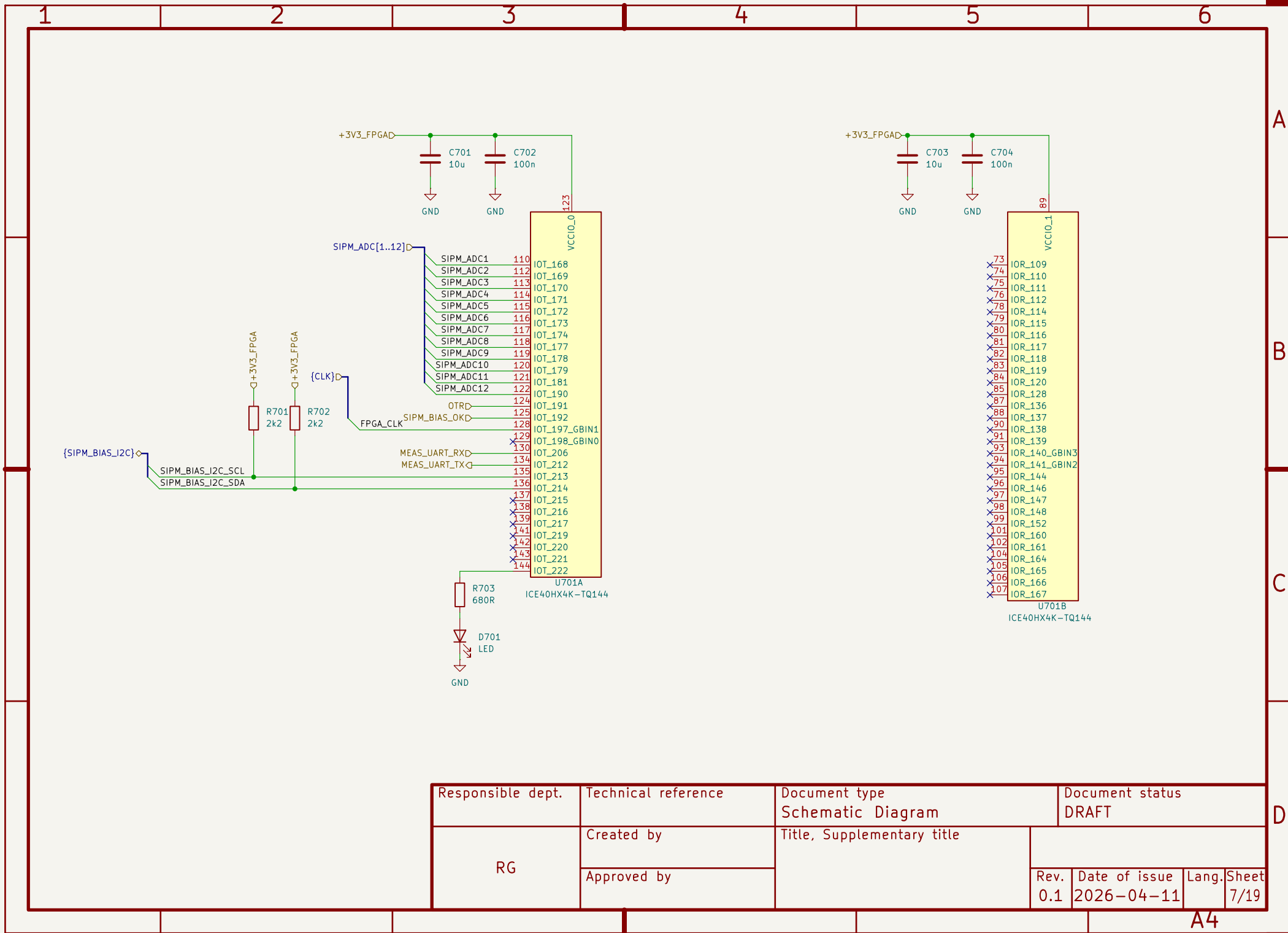
Label	Description
VCC_SIPM	28.89V–29.22V (+322.5mV over 15degC) SIPM bias
VCC_5V	Differential ADC Driver and its reference voltage
VCC_5V	Differential ADC Driver
VCC_2V5	TIA
VEE_2V5	TIA



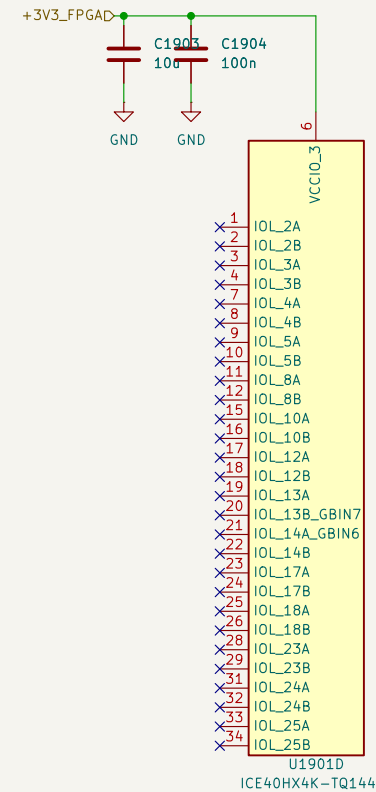
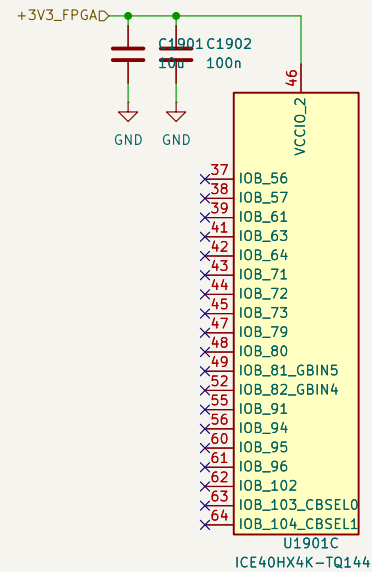
Responsible dept.	Technical reference	Document type	Document status
RG	Created by	Title, Supplementary title	DRAFT
	Approved by	TIA & Differential ADC Driver	Rev. 0.1
			Date of issue 2026-04-11
			Lang. 3/19



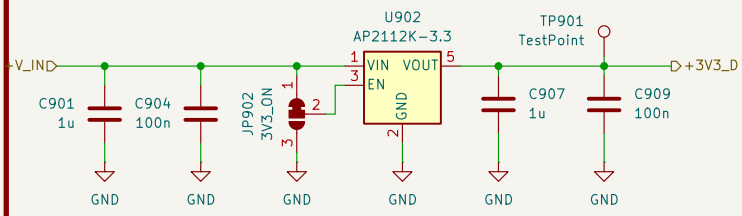
Responsible dept.	Technical reference		Document type		Document status		
			Schematic Diagram		DRAFT		
	Created by		Title, Supplementary title				
RG	Approved by		ICE40HX4K FPGA Core & I/O Banks		Rev.	Date of issue	Lang.
					0.1	2026-04-11	6/19



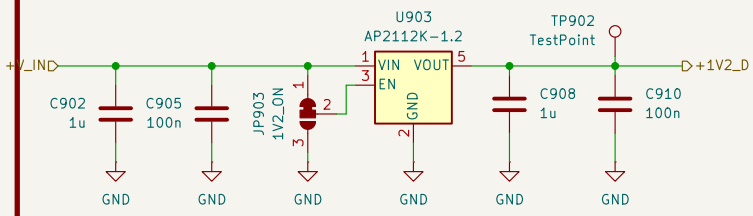
Responsible dept.	Technical reference	Document type	Document status
RG	Created by	Schematic Diagram	DRAFT
	Approved by	Title, Supplementary title	
		Rev.	Date of issue
		0.1	2026-04-11
		Lang.	Sheet
			7/19



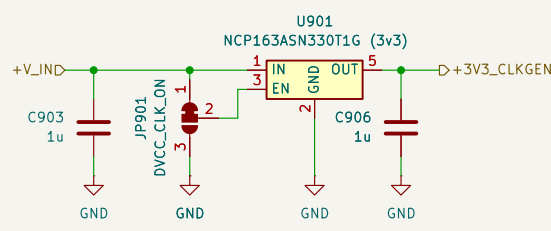
Responsible dept.	Technical reference	Document type Schematic Diagram	Document status DRAFT			
RG	Created by	Title, Supplementary title				
	Approved by		Rev. 0.1	Date of issue 2026-04-11	Lang.	Sheet 19/19



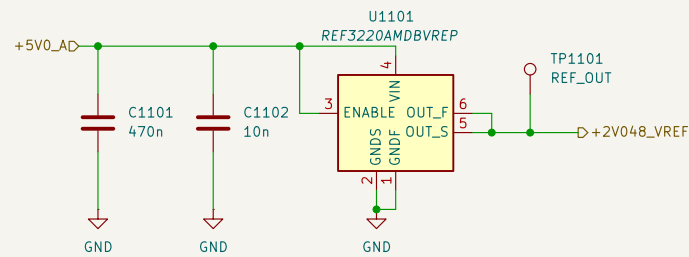
Label	Direction	Description
VCC	Input	Raw input voltage (6V)
VCC_SIPM	Output	30V for analog frontend bias
VCC_3V3	Output	3.3V for FPGA I/O, flash, ADC digital side
VCC_1V2	Output	1.2V for FPGA core
VCC_5V	Output	5V for ADC
VCC_2V5	Output	2V5 for amplifier
VEE_2V5	Output	-2V5 for amplifier
GND	Bidirectional	Ground



+2V5_D



Responsible dept.	Technical reference	Document type Schematic Diagram		Document status DRAFT			
RG	Created by	Title, Supplementary title Digital Power Regulators (3.3V, 1.2V)					
	Approved by			Rev. 0.1	Date of issue 2026-04-11	Lang.	Sheet 9/19



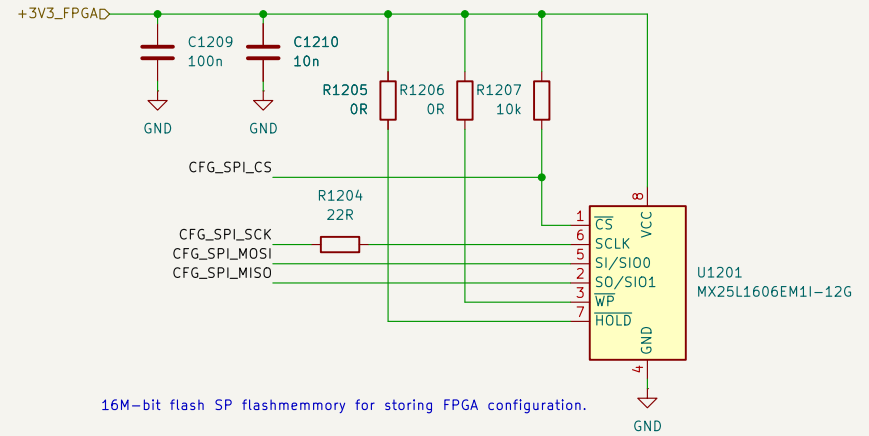
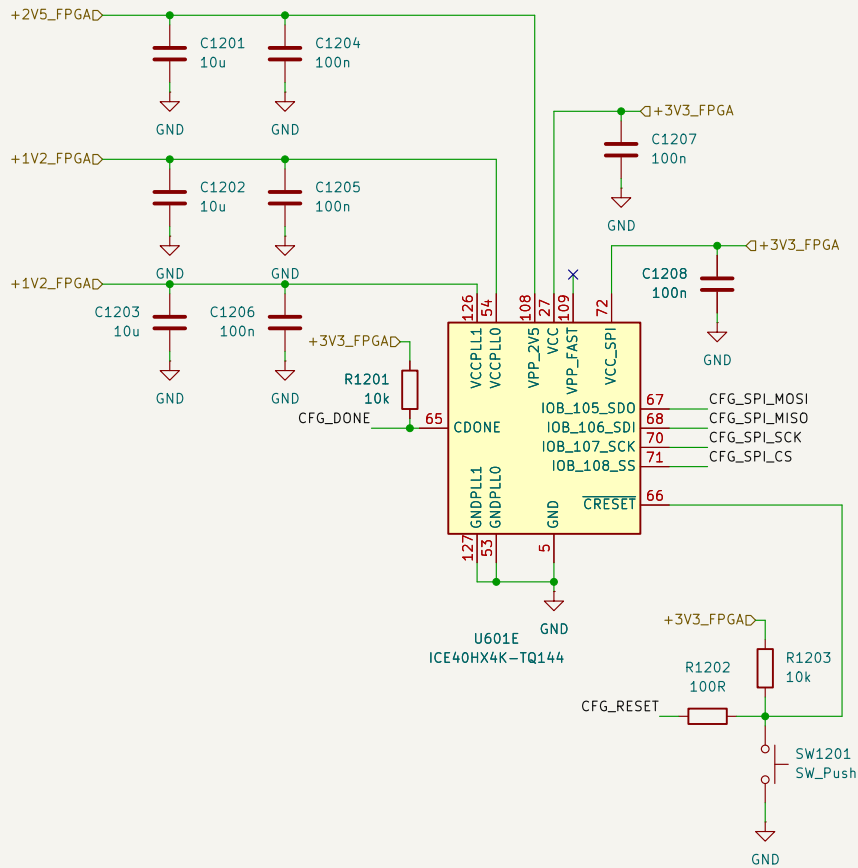
REF3220AMDBVREP is 2,048V precision voltage reference.

Place U13 (REF3220) physically close to the AD9226 ADC.

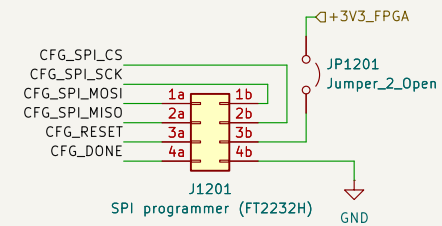
Kelvin Sense Connections:

- OUT_S (Sense+) must be routed as a dedicated trace directly to the ADC's V_REF pin.
- GND_S (Sense-) must be routed as a dedicated trace directly to the ADC's REF_COM pin.

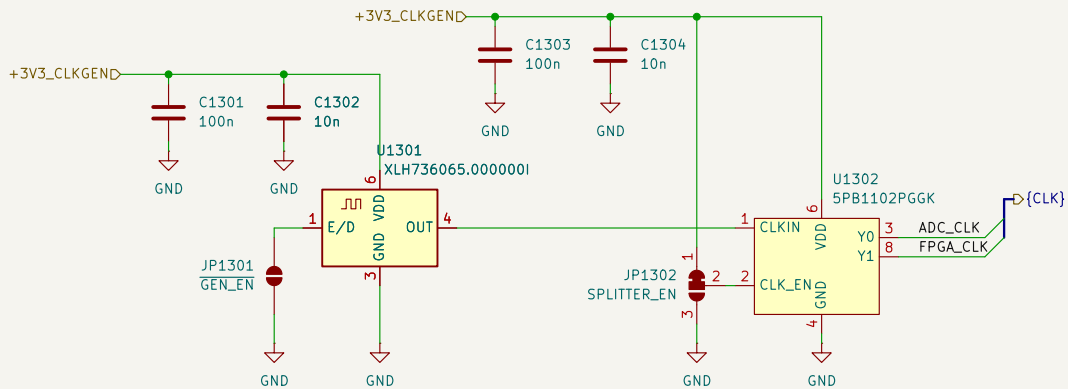
Responsible dept.	Technical reference	Document type Schematic Diagram		Document status DRAFT		
RG	Created by	Title, Supplementary title				
	Approved by		Rev. 0.1	Date of issue 2026-04-11	Lang.	Sheet 11/19



16M-bit flash SP flashmemory for storing FPGA configuration.

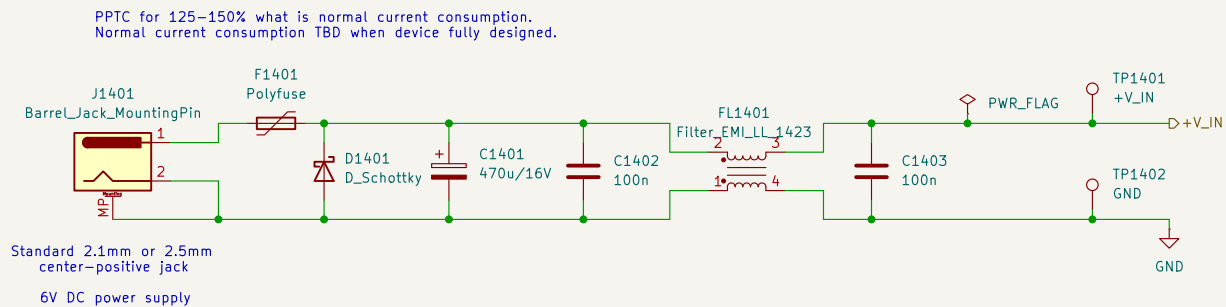


Responsible dept.	Technical reference	Document type	Document status
		Schematic Diagram	DRAFT
RG	Created by	Title, Supplementary title	
	Approved by	FPGA Configuration Memory – External 16Mb SPI Boot Flash	Rev. 0.1 Date of issue 2026-04-11 Lang. Sheet 12/19

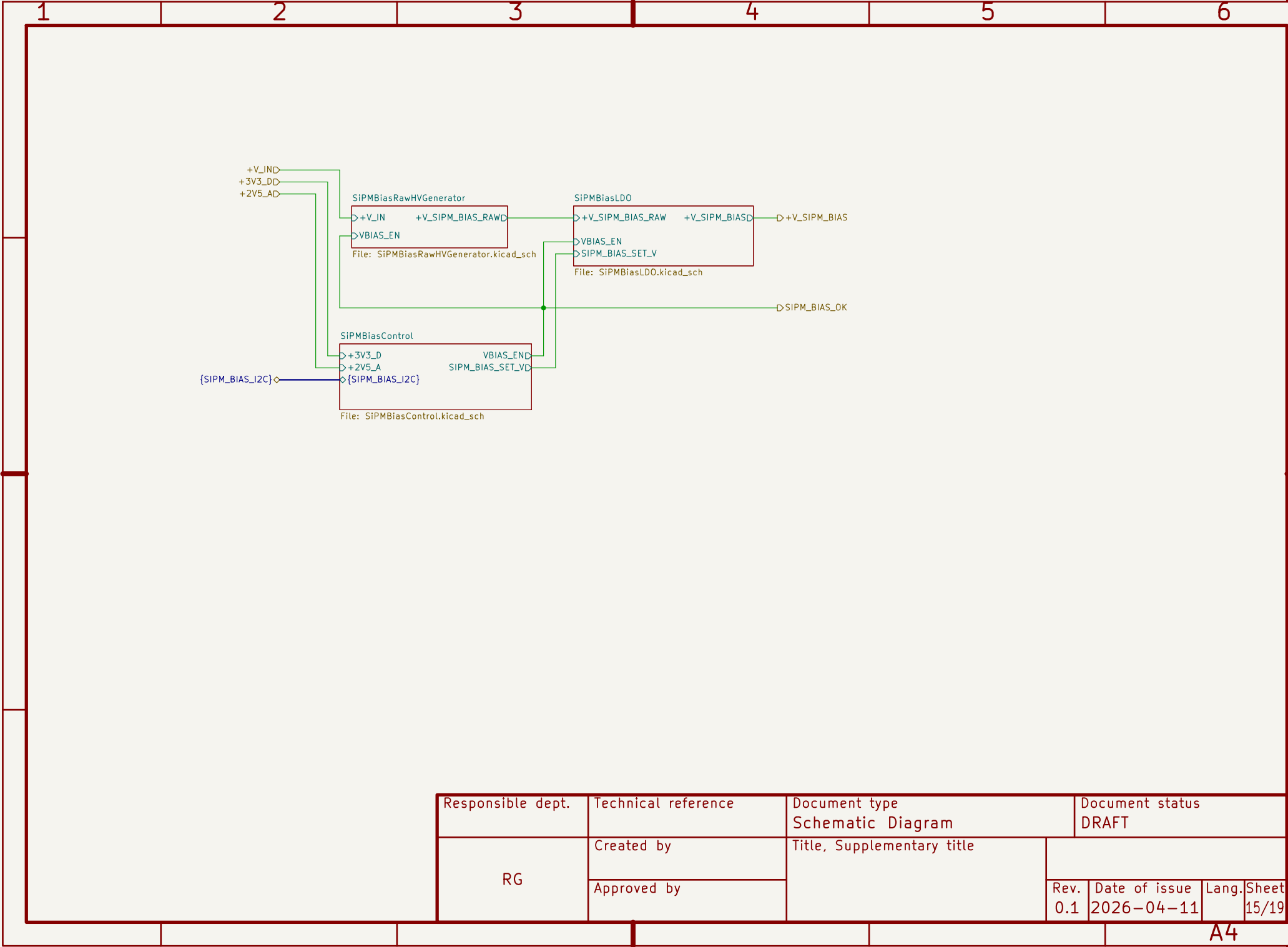


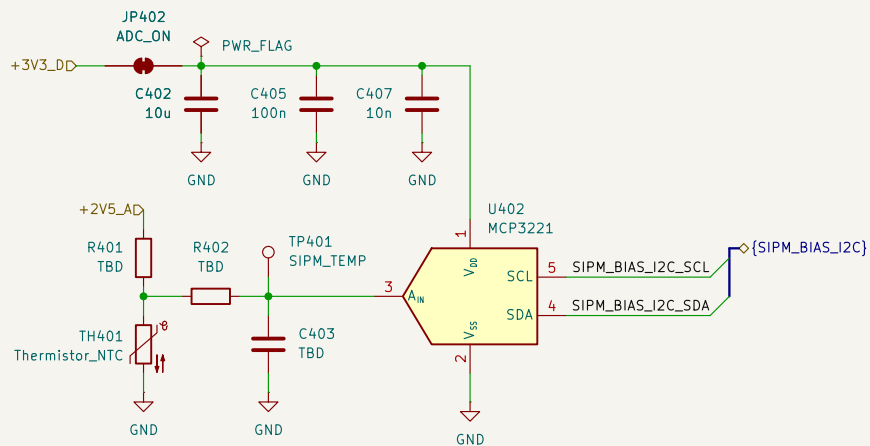
CLK is 65MHz 25ppm
5PB1102PGGK is 1:2 CLK splitter

Responsible dept.	Technical reference	Document type	Document status
		Schematic Diagram	DRAFT
RG	Created by	Title, Supplementary title System Clock – 65 MHz Oscillator and Dual Output Buffer	Rev. 0.1
	Approved by		
			Date of issue 2026-04-11
			Lang.
			Sheet 13/19

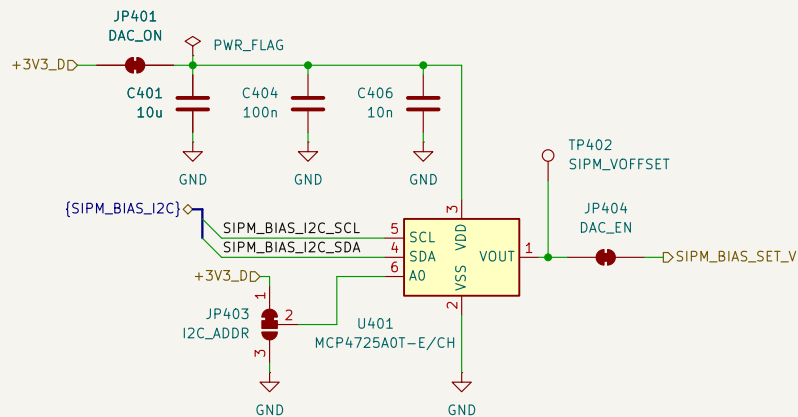


Responsible dept.	Technical reference	Document type	Document status
		Schematic Diagram	DRAFT
RG	Created by	Title, Supplementary title	
	Approved by	Barrel Jack Input and Preliminary Protection Interface	
		Rev. 0.1	Date of issue 2026-04-11
		Lang.	Sheet 14/19

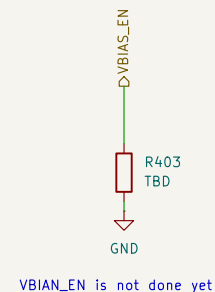




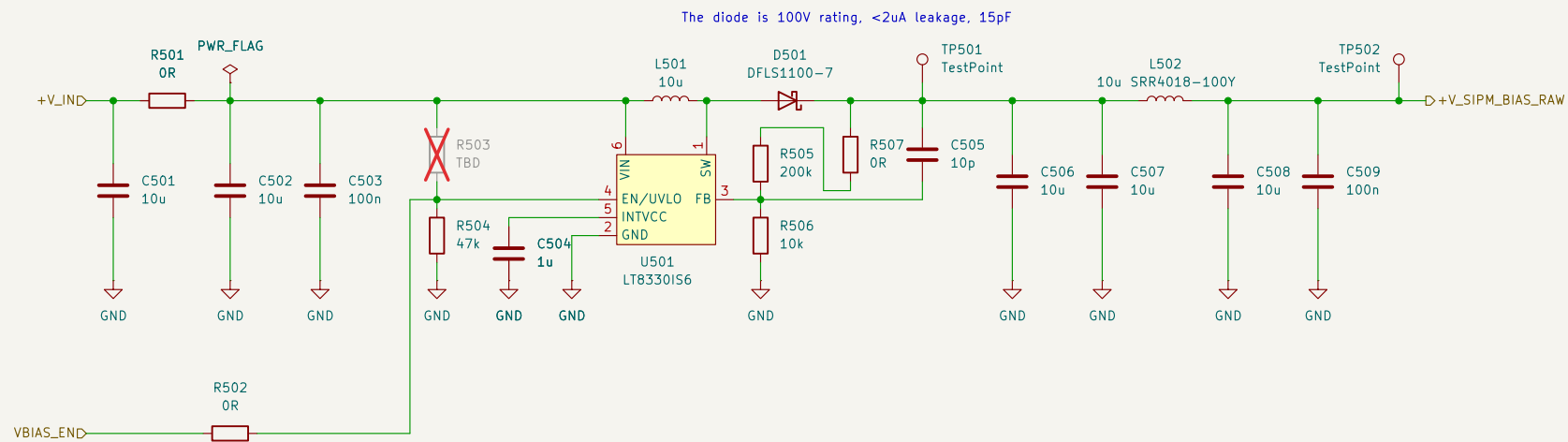
MCP3221A5T-E/OT: 12-bit I2C ADC with a fixed I2C address.



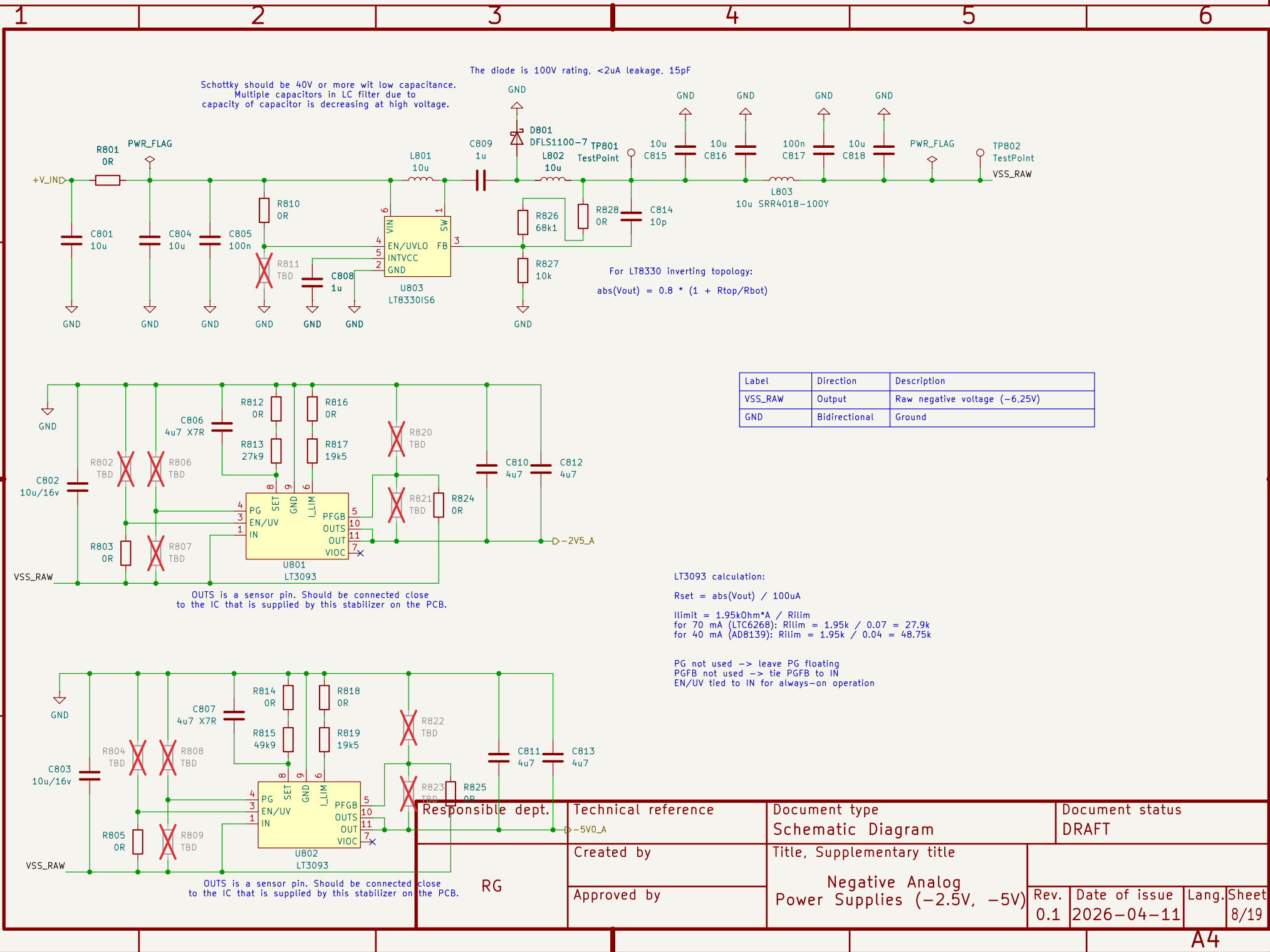
MCP4725A0T-E/CH is a 12-bit I2C DAC with an address that can be modified via the A0 pin.

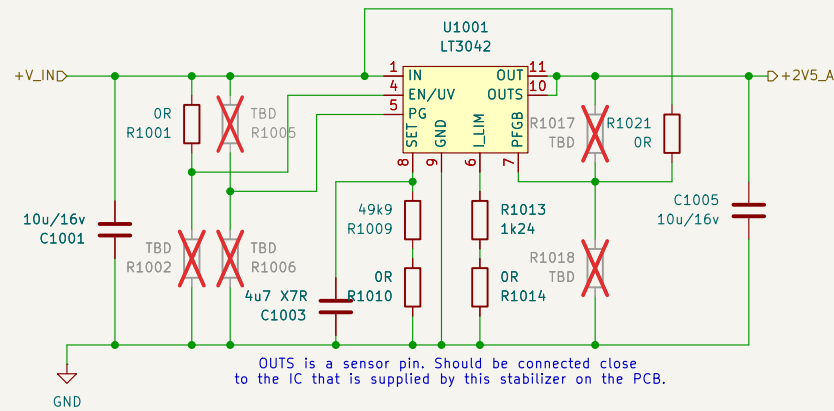


Responsible dept.	Technical reference	Document type	Document status
		Schematic Diagram	DRAFT
RG	Created by	Title, Supplementary title	
	Approved by	SIPM Bias Control and Temperature Compensation	
	REV.	Date of issue	Lang.
	0.1	2026-04-11	
			Sheet
			4/19



Responsible dept.	Technical reference	Document type	Document status
		Schematic Diagram	DRAFT
RG	Created by	Title, Supplementary title	
	Approved by		
		High-Voltage SiPM Bias Generator	Rev. 0.1 Date of issue 2026-04-11 Lang. Sheet 5/19





LT3042 calculation:

$$R_{set} = \text{abs}(V_{out}) / 100\mu A$$

$$I_{limit} = 125\text{mA} \cdot k\Omega\text{m} / R_{lim}$$

For:

$$\text{for } 60 \text{ mA: } R_{lim} = 125 / 60 = 2.08\text{k}$$

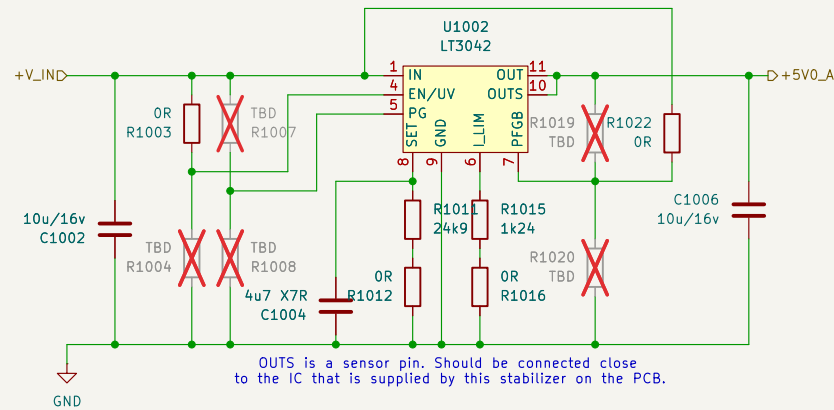
$$\text{for } 70 \text{ mA: } R_{lim} = 125 / 70 = 1.79\text{k}$$

$$\text{for } 100 \text{ mA: } R_{lim} = 125 / 100 = 1.24\text{k}$$

PG not used -> leave PG floating

PGFB not used -> tie PGFB to IN

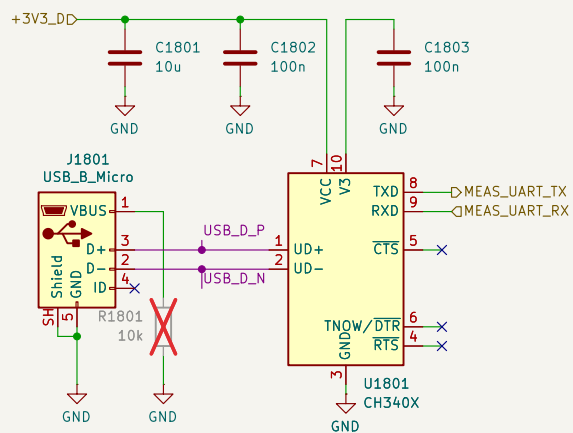
EN/UV tied to IN for always-on operation



HIGH QUALITY CAPS FOR SET PIN!!!

D+3V3_ADC

Responsible dept.	Technical reference	Document type Schematic Diagram		Document status DRAFT			
RG	Created by	Title, Supplementary title Positive Analog Power Supplies (2.5V, 5V)					
	Approved by			Rev. 0.1	Date of issue 2026-04-11	Lang.	Sheet 10/19



Responsible dept.	Technical reference	Document type Schematic Diagram		Document status DRAFT			
RG	Created by	Title, Supplementary title External Interfaces					
	Approved by			Rev. 0.1	Date of issue 2026-04-11	Lang.	Sheet 18/19